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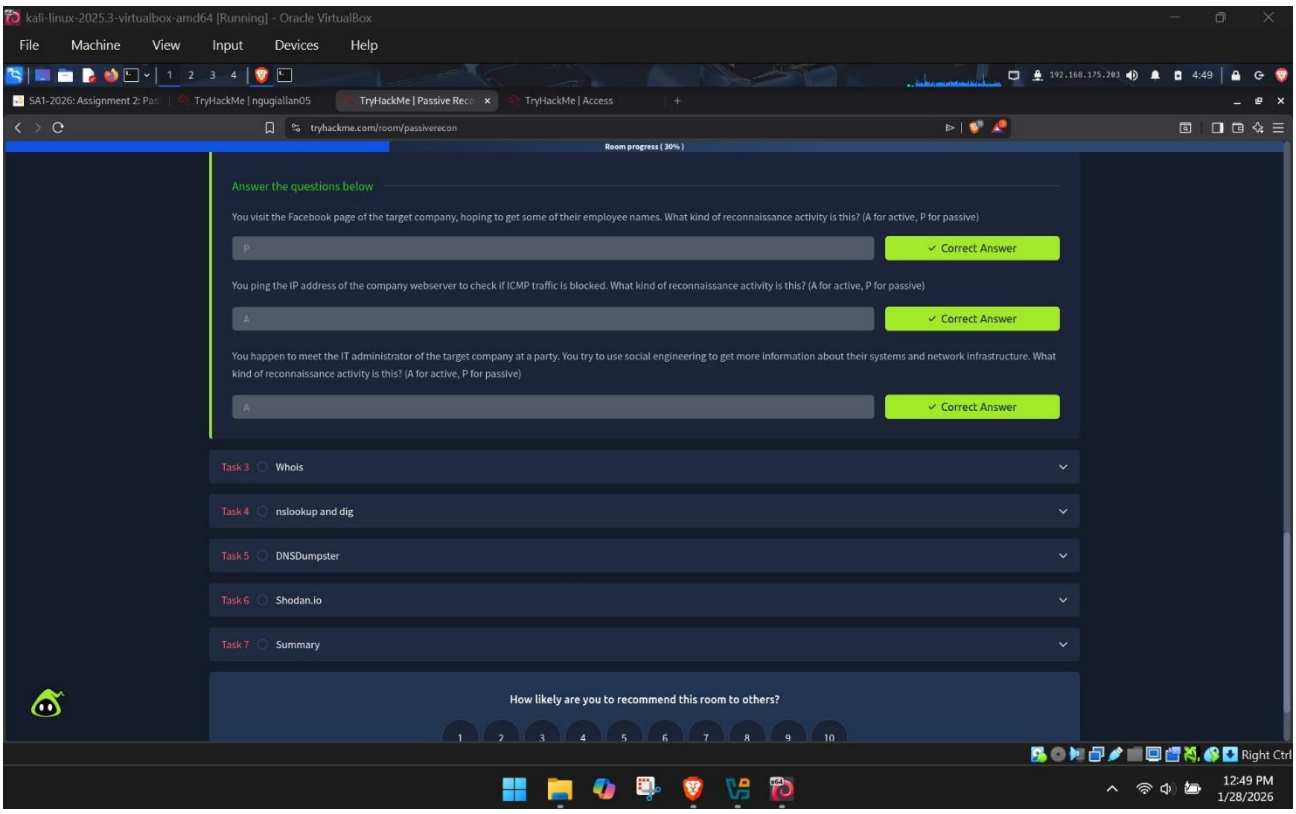
Introduction

This room introduced me to the inaugural section of the Network Security Module, which focuses on active and particularly, passive reconnaissance. The text explains how users can collect public records through non-intrusive command-line tools whois and nslookup and dig while also using external platforms DNSDumpster and Shodan.io to gather intelligence without informing their targets. The material establishes a stealthy base for information gathering through its use of publicly available data and basic networking requirements before advancing to direct scanning methods.

Task 2: Passive versus Active Recon

The task mainly focuses on distinguishing the difference between active and passive recon, giving practical examples in each.



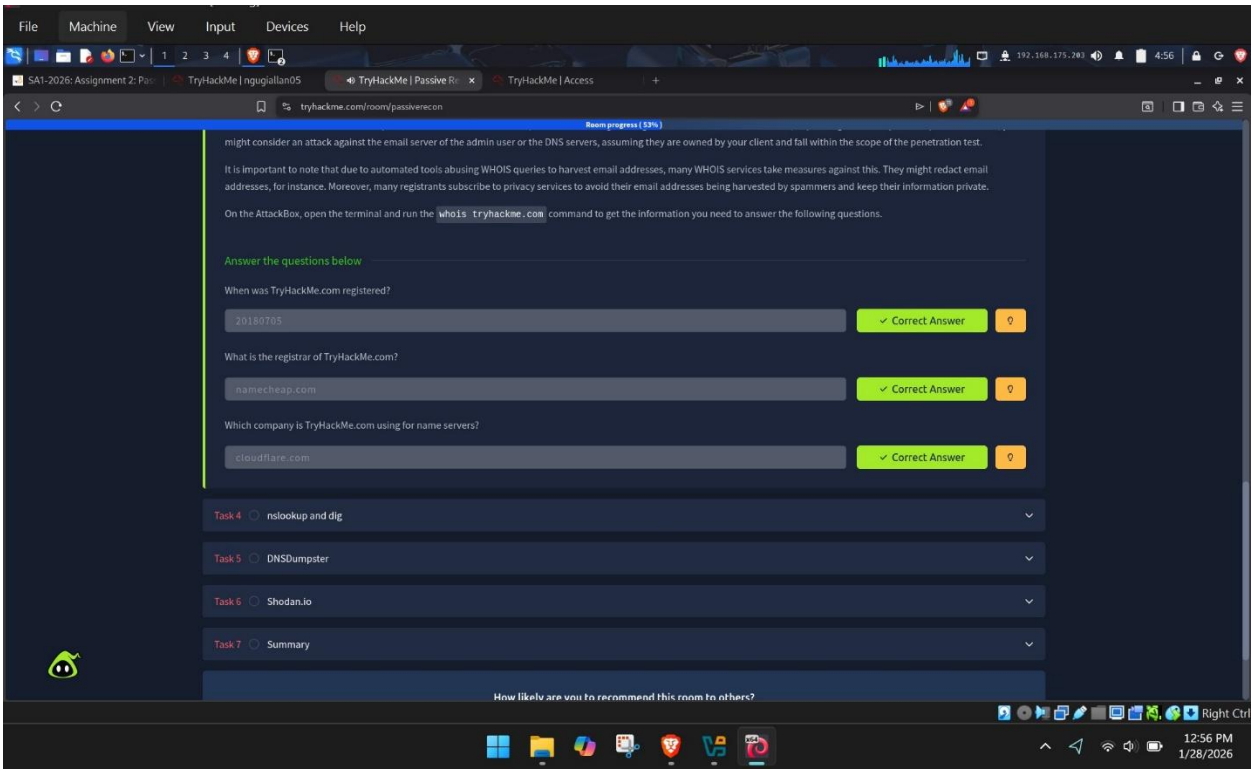
For question 1, the reason that is passive reconnaissance is that it relies on publicly available knowledge through Facebook; and therefore, there is no direct contact with the target.

For question 2, the reason it is active is because there is direct involvement with the target in a discrete manner.

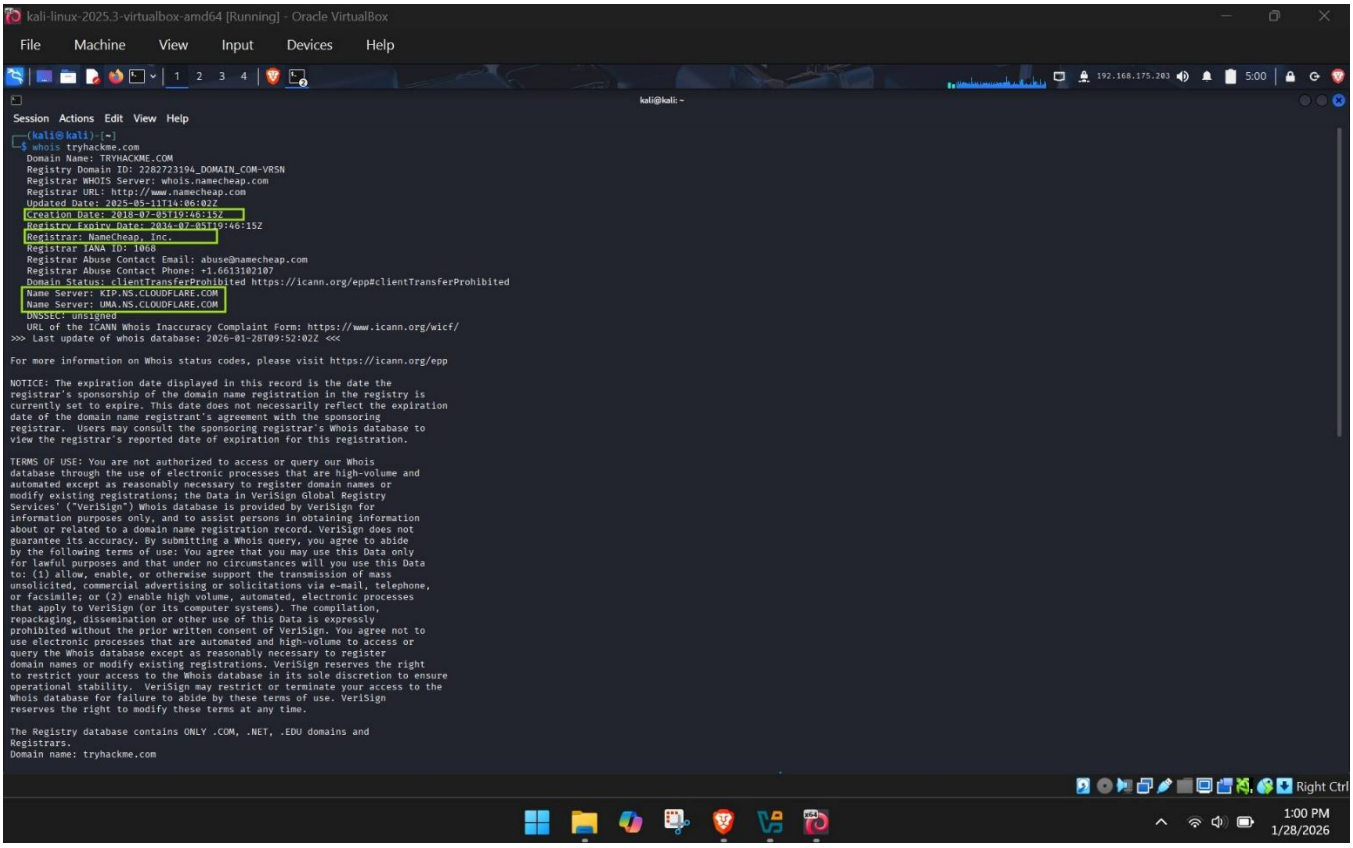
For question 3, social engineering is a form of direct reconnaissance since attempts to get information about the organization from an individual affiliated with the organization involved direct involvement.

Task 3: Whois

This task focuses on the whois command that is used to obtain information about a certain domain. An instance of a command ran using whois could be, ‘whois tryhackme.com’



The proof of answers is shown below:



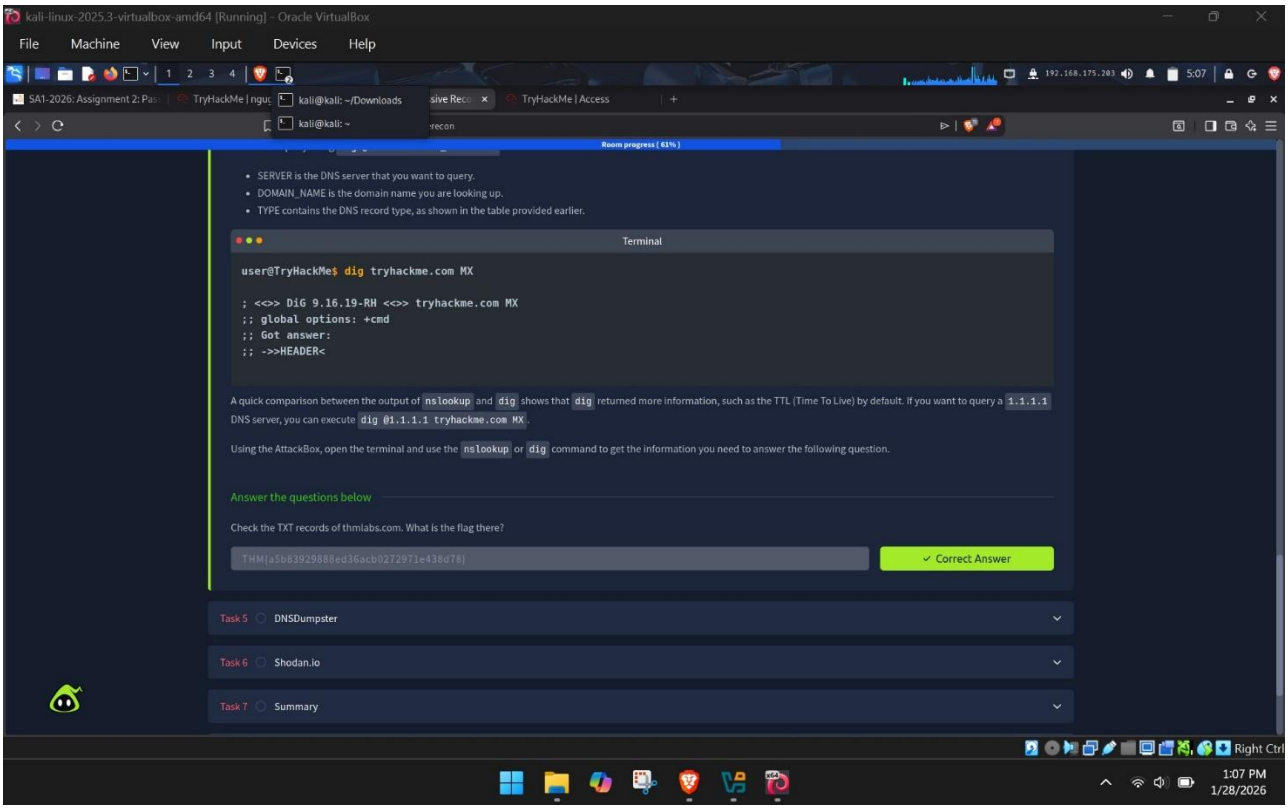
The first question is derived from the date of creation (first box)

The second question, which touches on who Try Hack Me's registrar is, is shown in the second box.

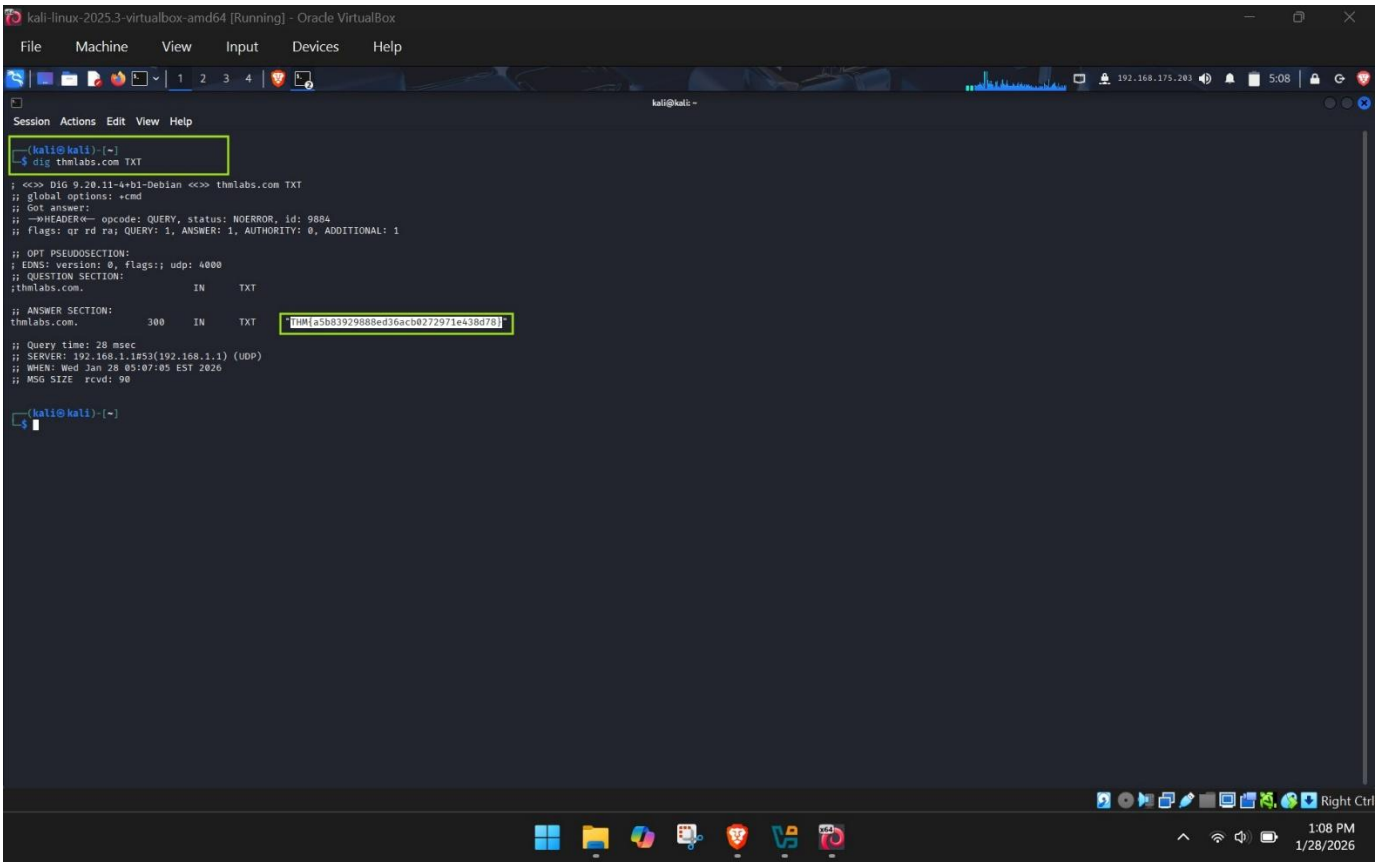
The final question touches on the company Try Hack Me uses for name servers, which is Cloudflare as shown in the very last box.

Task 4: nslookup and dig

This task concentrates on the tools command tools nslookup and dig. Nslookup and dig enable users to query DNS servers for domain name and IP address and technical record information. Nslookup provides users with a basic interface while dig delivers comprehensive technical information that users can use in their scripts.

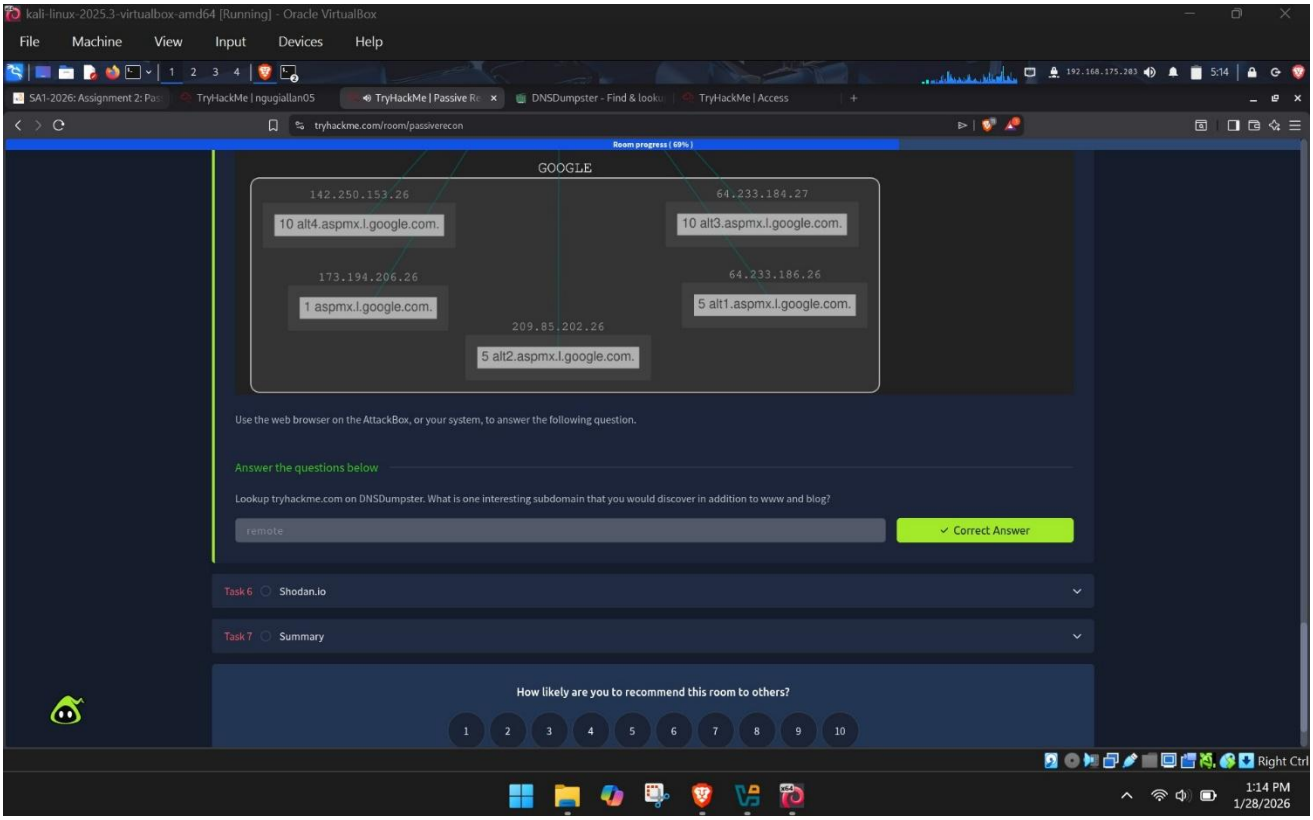


To get the flag, I used the dig command to obtain the flag. The command was dig thmlabs.com TXT. The TXT suggests that the file being a text record. The output is as shown below:

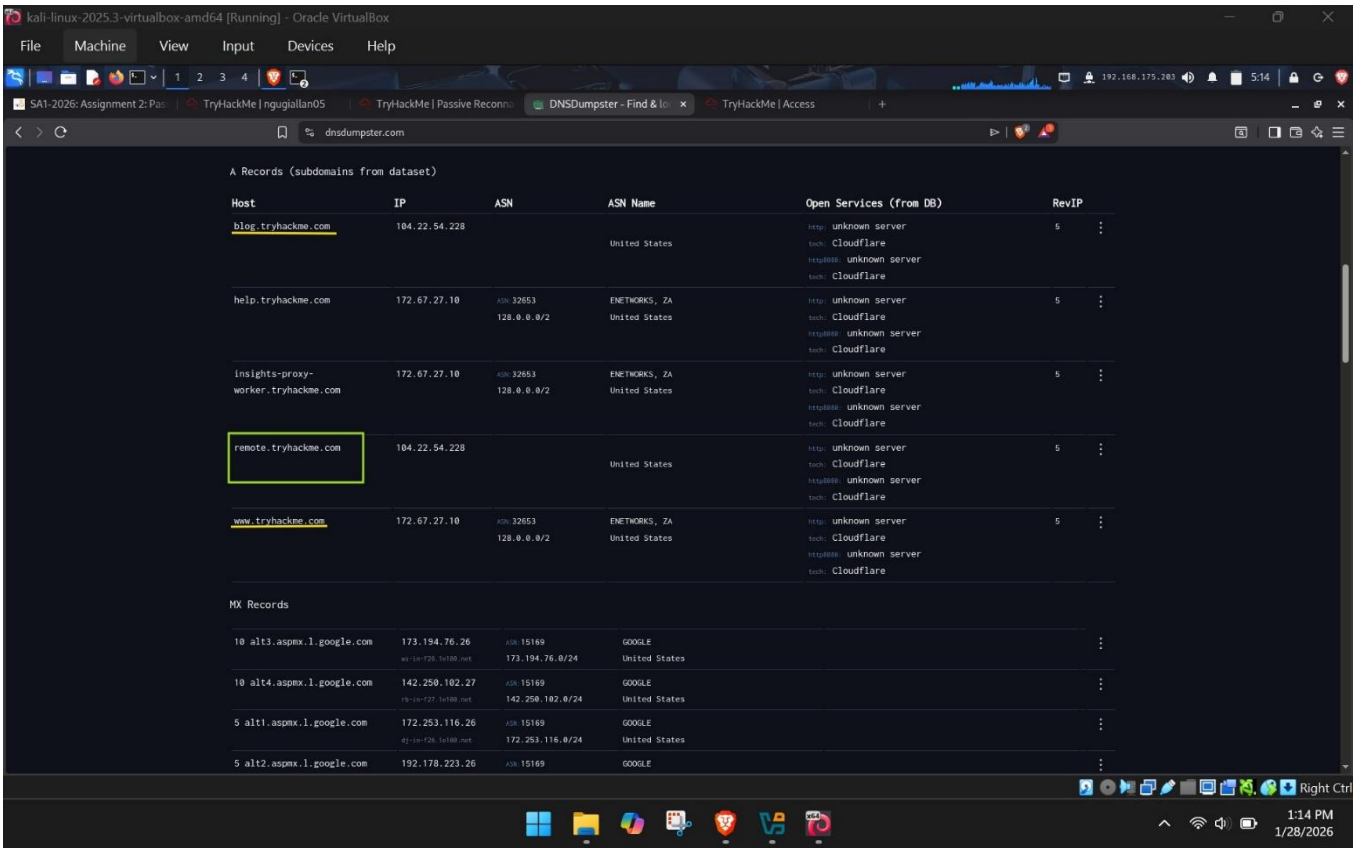


Task 5: DNSDumpster

The task touches on DNSDumpster which is an online service that serves as a complimentary tool which enables users to conduct passive reconnaissance for their domain research needs because it helps them identify and track all digital assets owned by an organization without needing to connect to the organization's servers. The system collects DNS records which are open to the public and uses them to identify subdomains and mail servers and IP addresses while displaying this information through graphical representations that show the target's network structure. Security professionals use it to find hidden subdomains which enables them to discover potential entry points while they gather information about the entire attackable area of a domain.

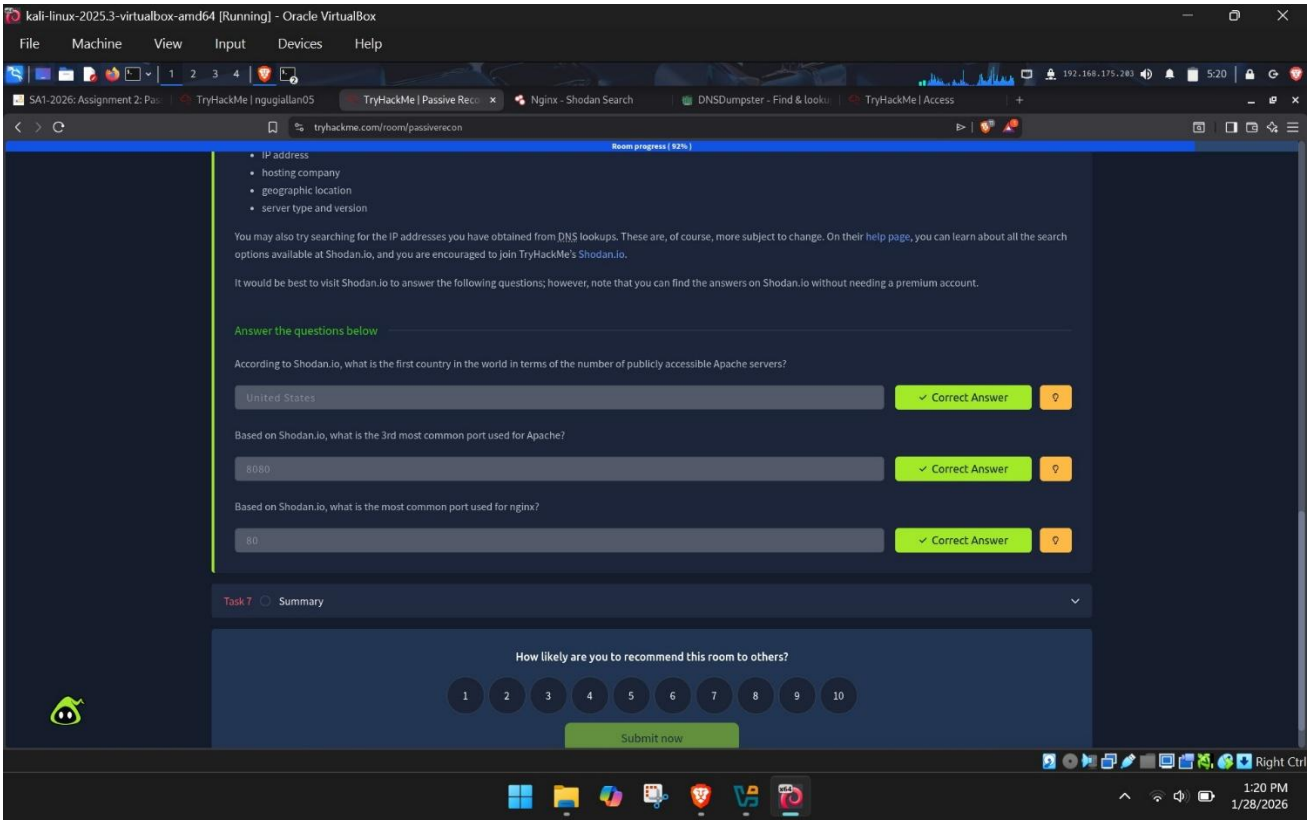


Once I searched the domain tryhackme.com on DNSDumpster, I was able to find the subdomain in addition to www and blog. This is shown below:

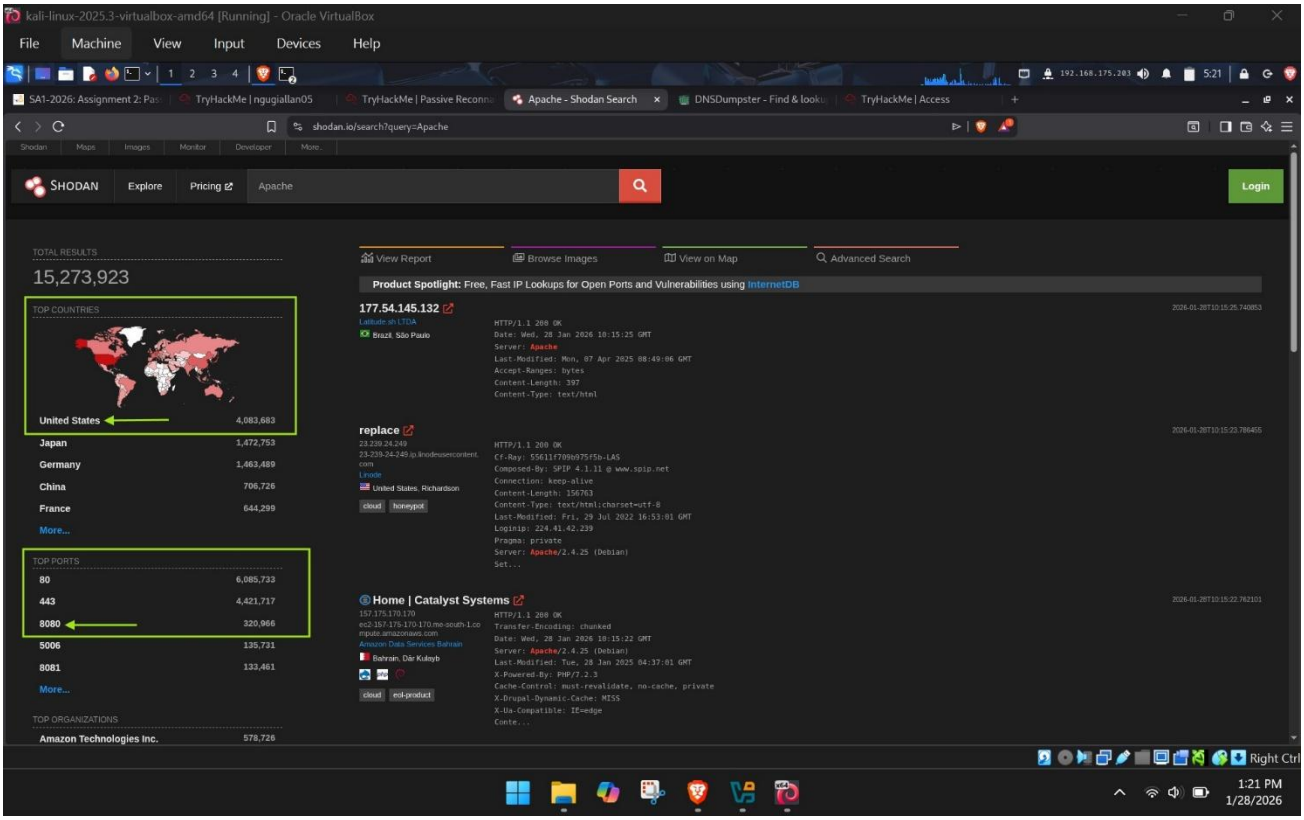


Task 6: Shodan.io

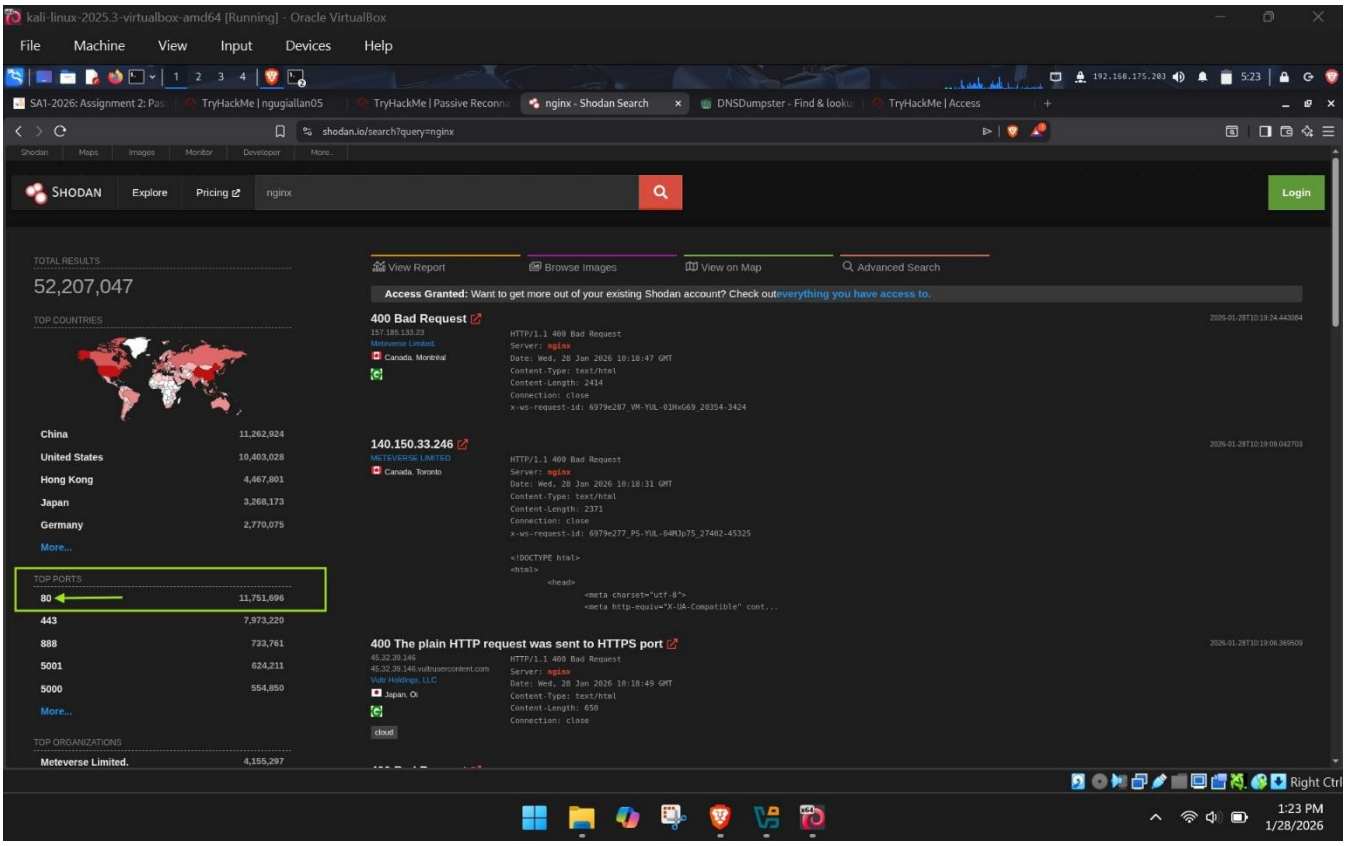
Shodan.io is a search engine which identifies internet-connected devices by collecting their metadata banners to display their assets which include servers and IoT devices. The system enables users to detect open ports and security weaknesses throughout the target network without needing to interact with its components. The system functions as a vital digital defence tool which enables security experts to conduct passive vulnerability assessments of their digital assets.



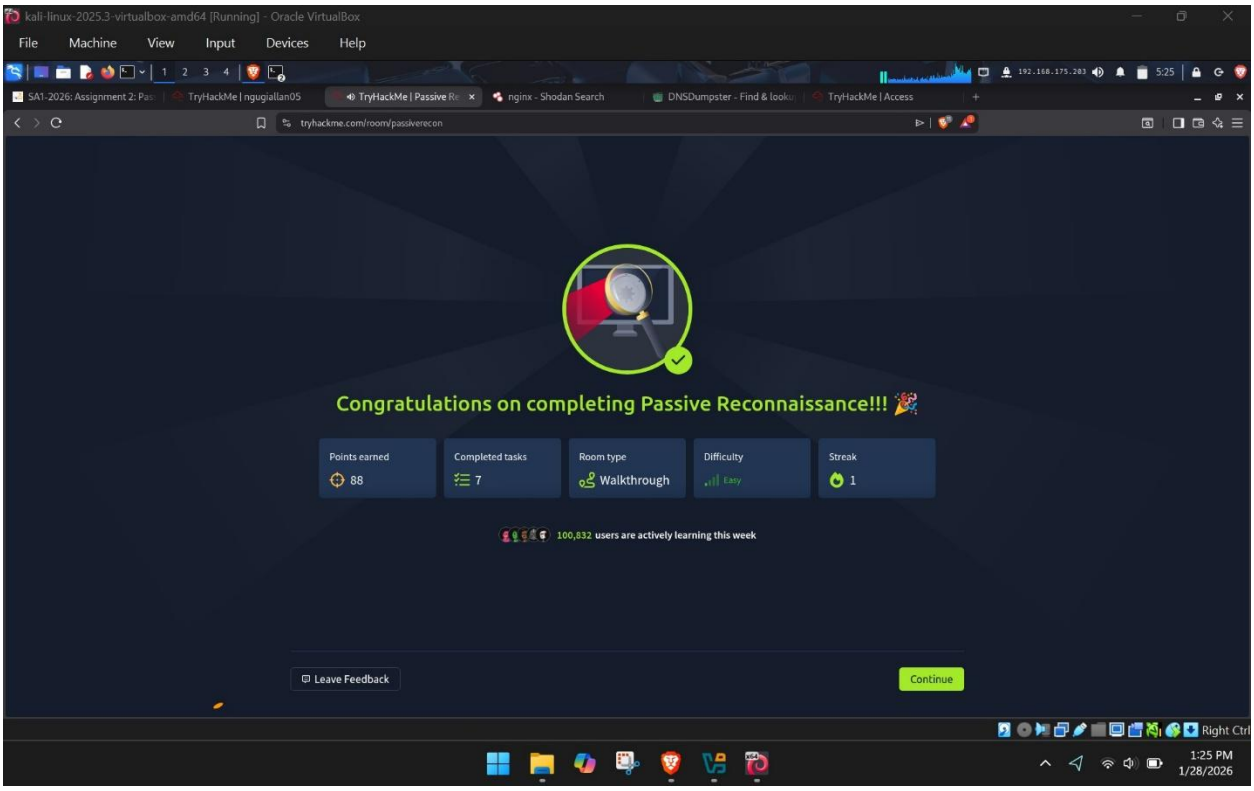
For the first and second questions, I searched on the Shodan's search bar Apache and was able to retrieve the information as shown below:



For the last question, I also used the search bar for nginx; and the output is as shown below:


kali-linux-2025.3-virtualbox-amd64 [Running] - Oracle VirtualBox
File
Machine
View
Input
Devices
Help
SA1-2026: Assignment 2: Pas
TryHackMe | ngugiallan05
TryHackMe | Passive Reconn
nginx - Shodan Search
DNSDumpster - Find & looku
TryHackMe | Access
shodan.io/search?query=nginx
SHODAN
Explore
Pricing
nginx
Login
TOTAL RESULTS
52,207,047
TOP COUNTRIES
China 11,262,924
United States 10,403,028
Hong Kong 4,467,801
Japan 3,268,173
Germany 2,770,075
More...
TOP PORTS
80 11,751,696
443 7,973,220
888 733,761
5001 624,211
5000 554,850
More...
TOP ORGANIZATIONS
Meteverse Limited. 4,155,297
View Report
Browse Images
View on Map
Advanced Search
Access Granted: Want to get more out of your existing Shodan account? Check out everything you have access to.
400 Bad Request
157.185.133.23
Meteverse Limited
Canada, Montréal
HTTP/1.1 400 Bad Request
Server: nginx
Date: Wed, 28 Jan 2026 10:18:47 GMT
Content-Type: text/html
Content-Length: 2414
Connection: close
x-ws-request-id: 6979e287_VM-YUL-01HxG69_20354-3424
140.150.33.246
METEVERSE LIMITED
Canada, Toronto
HTTP/1.1 400 Bad Request
Server: nginx
Date: Wed, 28 Jan 2026 10:18:31 GMT
Content-Type: text/html
Content-Length: 2371
Connection: close
x-ws-request-id: 6979e277_PS-YUL-04MJp75_27402-45325
<!DOCTYPE html>
<html>
<head>
<meta charset="utf-8">
<meta http-equiv="X-UA-Compatible" cont...
400 The plain HTTP request was sent to HTTPS port
45.32.39.146
45.32.39.146.vultrusercontent.com
Vultr Holdings, LLC
Japan, Ōi
HTTP/1.1 400 Bad Request
Server: nginx
Date: Wed, 28 Jan 2026 10:18:49 GMT
Content-Type: text/html
Content-Length: 650
Connection: close
cloud
Right Ctrl
1:23 PM
1/28/2026

Screenshot of Room Completion



Conclusion

The room introduced the aspects of active and passive reconnaissance, clearly distinguishing them, and also dived into the various tools and methodologies that one can use to carry out passive reconnaissance, which does not need direct involvement with the target. This enlightened me on the various tools that I did not know of, empowering my knowledge and skills in reconnaissance.

Link to Completed Module

https://tryhackme.com/p/ngugiallan05